

\*  $\ell^n$ : nowhere vanishing vect. field

$\rightarrow$  Congruence set of curves generated by the curves

$\rightarrow \ell^n \sim \tilde{\ell}^m \Leftrightarrow \exists f \in C^\infty(M)$  s.t.  $\ell^m = f \tilde{\ell}^m$

• Null congruence  $\ell^m \ell_m = 0$   $[\ell^m]$

• Geodesic congruence  $\ell^m \nabla_\mu \ell^m = 0$

Prop: Consider a NGL generated by  $\ell^m$  one can always find (locally)

a coord. system  $(U, r, y^A)$   $A=1,2$

s.t.  $\rho = \partial/\partial r$

$g \in \mathbb{R}$   
 $g^1 = 2, g^2 = \bar{2}$

(Newman-Unti Gauge)

$$ds^2 = V(U, r, y) du^2 + 2du dr + \gamma_{AB}(dy^A - U^A du)(dy^B - U^B du) + 2D_A(U, r, y^0) dy^A dr$$

\* Geodesic constraint:  $\partial_r D_A = 0$   $|\ell = \frac{\partial}{\partial r}$

$$\rightarrow \ell^m \nabla_\mu \ell^m = \partial_r (\Gamma_{\mu\sigma}^r \partial^\sigma)$$

$$\Gamma_{\mu\nu}^r = \frac{1}{2} g^{r\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$$

$\sigma = U$  or  $\sigma = A$

$$\Gamma_{\mu\nu}^r = \frac{1}{2} g^{rU} (\partial_\mu g_{U\nu} + \partial_\nu g_{U\mu} - \partial_U g_{\mu\nu}) + \frac{1}{2} g^{rA} (\partial_\mu g_{A\nu} + \partial_\nu g_{A\mu} - \partial_A g_{\mu\nu})$$

$$g^{rU} = 1$$

$$g^{rA} = \frac{1}{2} D^A$$

$$\Gamma_{rr}^r \rightarrow \frac{1}{2} (\partial_r g_{Ur} + \partial_r g_{Ur} - \partial_U g_{rr}) + \frac{1}{4} D^A (\partial_r D_A + \partial_r D_A - \partial_A) = \frac{1}{2} D^A \partial_r D_A$$

$$\therefore \partial_r (D^A \partial_r D_A) = \partial_r D^A \partial_r D_A + D^A \partial_r^2 D_A = 0$$

Vanishes if  $\partial_r D_A = 0$  as claimed above.

\* Remark:  $\ell^m g_{m\nu} dx^\nu = du + D_A dy^A$

proof:

$$\ell^\mu = 1 \text{ since } \ell = \ell^r \frac{\partial}{\partial r}$$

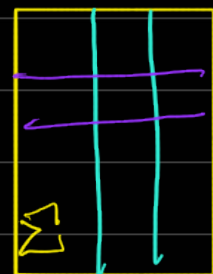
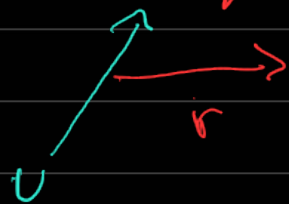
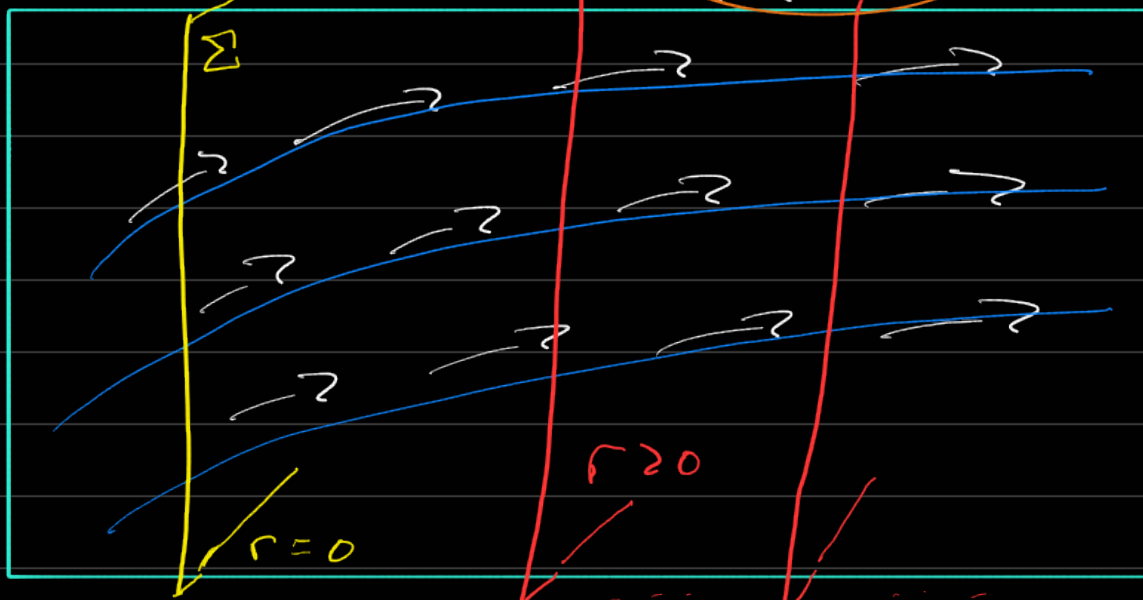
$$\ell^m g_{m\nu} dx^\nu = g_{r\nu} dx^\nu = g_{rU} du + g_{rA} dy^A$$

$$= 1 du + D_A dy^A$$

Definition: A NCC is a Hypersurface-orthogonal  
 if  $\exists$  a function  $U$  such that  $l^m g_{mn} dx^n = dU$

- $l^m$  really is orthogonal to the surface  $U = \text{const.}$   
 $U = \text{const.} \Rightarrow \nabla_m U$ : Normal to  $U = \text{const.}$   
 $dU = \nabla_m U dx^m$   
 and from above  $l^m g_{mn} dx^n = \nabla_m U dx^m$   
 or  $l^m g_{mn} = \nabla_n U$   
 which is normal to the surface.

$\Rightarrow P_A dy^A = 0 \Rightarrow * \textcircled{D_A = 0} *$   $l_{0n} = \text{congruence}$



\*  $l^m$  is along these constant  $U$  surfaces

\*  $U = \text{const.}$  are null Hypersurfaces

\*  $l^m$  is both tangent and normal to  $U = \text{const.}$   
 $\leadsto$  "non-twisting" congruences

Prop:  $\ell^\mu$  is hypersurface orthogonal

$\Leftrightarrow \ell_{[\mu} \partial_\nu \ell_{\rho]} = 0$  (Frobenius thm)  
 $\Rightarrow$  This is a constant on the type of vector field we are considering.

Null tetrad:  $(\ell^\mu, m^\mu, \bar{m}^\mu, n^\mu)$  frame

here  $\ell^\mu, n^\mu \in T_p M$

$$\begin{aligned} m^\mu &= m_1^\mu + i m_2^\mu \in \mathbb{C} \otimes T_p M \\ \bar{m}^\mu &= m_1^\mu - i m_2^\mu \in \mathbb{C} \otimes T_p M \end{aligned} \quad \left| \begin{array}{l} m_1, m_2 \in T_p M \end{array} \right.$$

is such that all contractions wrt.  $g_{\mu\nu}$  are zero but

$$\begin{aligned} \ell^\mu n^\sigma g_{\mu\sigma} &= -1 \quad (\text{timelike}) \\ \bar{m}^\nu m^\sigma g_{\nu\sigma} &= 1 \quad (\text{spacelike}) \end{aligned}$$

$$\Rightarrow \ell^2 = n^2 = m^2 = \bar{m}^2 = 0$$

and  $m_\alpha m_{\beta\mu} = \delta_{\alpha\beta} \left(\frac{1}{2}\right)^2$ ,  $\alpha, \beta \in \{1, 2\}$

$\Rightarrow$  Would this imply  $n^\mu$  is parallel to  $\ell^\mu$ ?

$$(m_1^\nu - i m_2^\nu)(m_1^\sigma + i m_2^\sigma) g_{\nu\sigma} = 1$$

$$m_1^\nu m_1^\sigma g_{\nu\sigma} + m_2^\nu m_2^\sigma g_{\nu\sigma} = 1$$

So  $m_\alpha^\mu m_{\beta\mu} = \frac{1}{2} \delta_{\alpha\beta}$  in order to keep the I-normalization.

$$\bullet m^2 = (m_1^\mu + i m_2^\mu)(m_{1\mu} + i m_{2\mu}) \stackrel{!}{=} 0$$

$$= m_1^2 - m_2^2 + 2i m_1 m_2$$

$$\bar{m}^2 = m_1^2 - m_2^2 - 2i m_1 m_2 \stackrel{!}{=} 0$$

Prop: Let  $\ell^\mu$  generate a NCC. Let  $(\ell^\mu, m^\mu, \bar{m}^\mu, n^\mu)$  be an adapted null tetrad. Then any other null tetrad  $(\ell^\mu, \hat{m}^\mu, \hat{\bar{m}}^\mu, \hat{n}^\mu)$  must be related by

$$\hat{m}^\mu = m^\mu + f(x) \ell^\mu$$

$$\hat{n}^\mu = n^\mu + f(x) m^\mu + \bar{f}(x) \bar{m}^\mu + \frac{1}{2} |f|^2 \ell^\mu$$

for any function  $f \in C(M)$

$$\ast \hat{\bar{m}}^\mu = \bar{m}^\mu + \bar{f}(x) \ell^\mu = (\bar{m}_1^\mu + \bar{f}(x) \ell^\mu) + i m_2^\mu$$

\* Note that the above  $\hat{m}^\mu, \hat{n}^\mu$  are symmetry transformations

\* The vector  $\ell^\mu$  is the same

$$g_{\mu\nu} = -\ell_\mu \ell_\nu + m_\mu \bar{m}_\nu$$

(i) Boosts  $\ell'^\mu = f(x) \ell^\mu$

which is simply a reparametrization of the congruence:

$$\ell^\mu = \frac{dx^\mu}{d\lambda} \quad \lambda \rightarrow \lambda'$$

$$\ell'^\mu = \frac{dx^\mu}{d\lambda'} = \frac{d\lambda}{d\lambda'} \frac{dx^\mu}{d\lambda} = \frac{d\lambda(\lambda')}{d\lambda'} \ell^\mu$$

$$\Rightarrow f(x) = d\lambda(\lambda')/d\lambda'$$

$$\Rightarrow n'^\mu = f^{-1}(\lambda) n^\mu$$

(ii) Spin (Spatial rotation)  $SO(2)$

$m_1$  and  $m_2$  span a Euclidean subspace

$$\begin{aligned} \tilde{m}_1 &= \cos(\theta) m_1 + \sin(\theta) m_2 \\ \tilde{m}_2 &= -\sin(\theta) m_1 + \cos(\theta) m_2 \end{aligned} \quad \begin{pmatrix} C & S \\ -S & C \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$$

$$\tilde{m} = e^{i\theta} m$$

(iii) Null rotations about  $\ell^\mu$

(a)  $\ell'^\mu = \ell^\mu$

(b)  $m'^\mu = \alpha \ell^\mu + \beta n^\mu + \gamma m^\mu + \delta \bar{m}^\mu$

$$\alpha, \delta \in \mathbb{C} \otimes \mathbb{C}(\mathcal{U})$$

Since  $\ell^2 = \ell \cdot m = \ell \cdot \bar{m} = 0$

$$\ell \cdot n = -1$$

$$\ell^\mu m'_\mu = -\beta \stackrel{!}{=} 0$$

$$\begin{aligned} \beta &= 0 \\ (m')^2 &\stackrel{!}{=} 0 \end{aligned}$$

$$= 2\gamma\delta$$

$$\gamma = 0 \quad \text{or} \quad \delta = 0$$

$\Rightarrow \gamma = 0$ : Exchanges  $m \leftrightarrow \bar{m}$ : reverses orientation of 2D E (excluded)

$$\Rightarrow \delta = 0$$

$$m'^\mu = \alpha \ell^\mu + \gamma m^\mu$$

$$\bar{m}'_\mu m^\mu = \gamma \bar{\gamma} = 1 \Rightarrow |\gamma| = 1$$

$\gamma = e^{i\theta}$ : spin transformation (Gauge)

Choose  $\gamma = 1$

$$m'^\mu = m^\mu + f(x)l^\mu$$

as claimed

(c) Let  $n' = n + \alpha m + \bar{\alpha} \bar{m} + \beta l$

similarly we find  $\alpha = \bar{f}$ ,  $\beta = f\bar{f}$

$\therefore$

Euclidean shear:  $l'^\mu = l^\mu$

$$m' = m + fl$$

$$\bar{m}' = \bar{m} + \bar{f}l$$

$$n' = n + \bar{f}m + f\bar{m} + \frac{1}{2}|f|^2 l$$

Corollary: Let  $(M, g)$  be a vacuum solution to Ein. eq. with  $\Lambda = 0$ . Let  $l^\mu$  generate a NCC. Then

$$\psi_0 = R_{\mu\nu\rho\sigma} l^\mu m^\nu l^\rho \bar{m}^\sigma \in \mathbb{C}$$

is an invariant of the NCC (Newman Penrose coeff.)

$\hookrightarrow$  does not depend on choice of  $m$

\*

$$\psi_1 := R_{\mu\nu\rho\sigma} l^\mu m^\nu l^\rho n^\sigma \in \mathbb{C}$$

In general,  $\psi_1$  is not invariant.

If  $\psi_0 = 0$ , then  $\psi_1$  becomes invariant of the NCC.

If  $\psi_0 = \psi_1 = 0$ , then

$$\psi_2 := R_{\mu\nu\rho\sigma} l^\mu m^\nu \bar{m}^\rho n^\sigma$$

is invariant

If  $\psi_0 = \psi_1 = \psi_2 = 0$ , then  $\psi_3$  is inv.

...

\* If  $\psi_4 = 0$  too then  $M$  is flat.

\*

\*

$\{\psi_0, \psi_1, \dots, \psi_4\}$ : 5 independent  $\mathbb{C}$  components which define the (Gauged) Weyl tensor

\*



## Sachs Peeling:

\* Can show that  $\tilde{C}_{abc}{}^d$  must vanish at  $\mathcal{I}^+$

$$\gamma(\lambda), \quad \gamma(0) = p_{\mu M}, \quad \lim_{\lambda \rightarrow \infty} \gamma(\lambda) = q \in \mathcal{I}^+$$

$$\left\{ \begin{array}{l} \therefore C_{abcd} = C_{abcd}^{(1)}/\lambda + C_{abcd}^{(2)}/\lambda^2 + C_{abcd}^{(3)}/\lambda^3 + C_{abcd}^{(4)}/\lambda^4 + \mathcal{O}(\frac{1}{\lambda}) \\ (i): \text{Bounded in } \tilde{M} \end{array} \right.$$

\* Sachs Idea: characterize on Asym flat S.T. by rewriting

$$\psi_0 \sim r^{-5} \psi_0^o + \mathcal{O}(r^{-4})$$

$$\psi_1 \sim r^{-4} \psi_1^o + \mathcal{O}(r^{-3})$$

$$\psi_2 \sim r^{-3} \psi_2^o + \mathcal{O}(r^{-2})$$

$\vdots$

$$\psi_4 \sim r^{-1} \boxed{\psi_4^o} + \mathcal{O}(r^0)$$

$\hookrightarrow$  slowest falloff: gravitational field

50  $\rightarrow$  10 R components of  $C_{abc}{}^d$

\* Note:

$$\begin{aligned} C_{\mu\nu\rho\sigma} l^\mu m^\nu \bar{l}^\rho \bar{m}^\sigma &= C_{\mu\nu\rho\sigma} l^\mu \bar{l}^\rho (m^\nu \bar{m}^\sigma + \bar{m}^\nu m^\sigma) \\ &\approx C_{\mu\nu\rho\sigma} l^\mu \bar{l}^\rho (g^{\nu\sigma}) = 0 \end{aligned}$$

## An aside on NGC: (Wald chp 9)

\* Singularities: Use "holes" left behind:  $\exists$  geodesics which are inextendible in at least one direction but have only a finite range of affine parameters

$\hookrightarrow$  Incomplete geodesics:

\* A spacetime is singular if it possesses at least one incomplete geodesic

\*  $\mathcal{O} \subset M$ .

A congruence in  $\mathcal{O}$  is a family of curves s.t. through  $\forall p \in \mathcal{O}$ , there passes precisely one curve in this family.  $\{\gamma_a\}_a \Rightarrow \bigcap_a \gamma_a = \emptyset$ ?

(i) Consider a congruence of timelike geodesics parameterized by proper time  $\tau$   
 $\{\xi^a\}$   $\xi^a \xi_a = -1$

Define:  $B_{ab} := \nabla_b \xi_a \leadsto d\xi \leadsto \nabla_a \xi_b$  is Normal to  $\{\gamma_a\}$

•  $B$  is purely "spatial":

$$\hookrightarrow B_{ab} \xi^a = B_{ab} \xi^b = 0$$

$$\xi^a \nabla_b \xi_a = \xi^b \nabla_b \xi_a = 0$$

• First term:  $\nabla_b (\xi^a \xi_a) = 0$

• Second term: Geodesic: i.e. timelike geodesic.

\* Consider a 1-parameter subgroup of geodesics in the congruence  $\gamma_s(\tau)$

Let  $\eta^a$  be the orthogonal deviation vector from  $\gamma_0$  for this subfamily

•  $T^a = (\partial/\partial \tau)^a$

(s,t) define coordinates of

$$\Sigma = \text{Span}\{\gamma_s(\tau)\}$$

• Geodesics:  $\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$

$$T^\mu = \frac{dx^\mu}{d\tau} : T^s = \frac{ds}{d\tau} = 0 \quad T_t = 1 \Rightarrow T = (1)^a \frac{\partial}{\partial \tau}$$

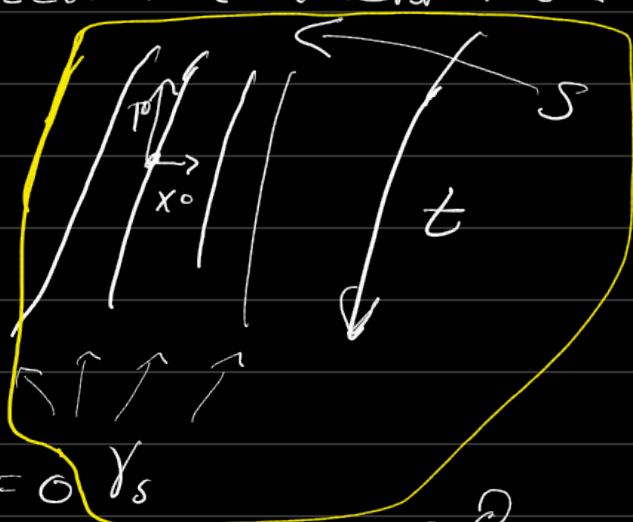
$$\Rightarrow T^a \nabla_a T^b = 0$$

\*  $\eta^a = (\partial/\partial s)^a$ : represents the displacement to an infinitesimally nearby geodesic (Deviation vector)

Note

$$(\mathcal{L}_\xi \eta)^a = [\xi, \eta]^a$$

$$= (\xi^\mu \partial_\mu (\eta^\nu \partial_\nu) - \eta^\nu \partial_\nu (\xi^\mu \partial_\mu))$$



$$\begin{aligned}
&= \xi^\mu (\partial_\mu n^\nu) \partial_\nu + \xi^\mu n^\nu \partial_\mu \partial_\nu \\
&\quad - n^\nu \xi^\sigma \partial_\nu \partial_\sigma - n^\nu (\partial_\nu \xi^\sigma) \partial_\sigma \\
&= [\xi^\sigma \partial_\sigma (n^\nu) \partial_\nu - n^\nu \partial_\nu (\xi^\sigma) \partial_\sigma]^a \\
&= \xi^\sigma \partial_\sigma n^a - n^\nu \partial_\nu \xi^a \\
&= \xi^b \nabla_b n^a - n^b \nabla_b \xi^a = 0 \\
&\therefore \boxed{\xi^b \nabla_b n^a = n^b B^a_b}
\end{aligned}$$

$\Rightarrow B^a_b$ : Measures the failure of  $n^a$  to be parallelly transported (along the congruence)

• Observer on  $\gamma_0$  would see nearby geodesics as stretched/rotated by  $B^a_b$

Spatial metric:

$$\boxed{h_{ab} := g_{ab} + \xi_a \xi_b}$$

• Note

$$\xi^a \xi^b h_{ab} = -1 + 1 = 0$$

Thus

$$h^{ac} h_{cb} = (g^{ac} + \xi^a \xi^c) h_{cb} = g^{ac} h_{cb}$$

$$\boxed{h^a_b = g^{ac} h_{cb}}$$

$\hookrightarrow$  Projection operator onto the subspace of the tangent space perpendicular to  $\xi^a$ .

$$\Theta \equiv B^a_b h_{ab} \quad (\text{Expansion})$$

$$\sigma_{ab} \equiv B_{(ab)} - \frac{1}{3} \Theta h_{ab} \quad (\text{Shear})$$

$$\omega_{ab} \equiv B_{[ab]} \quad (\text{Twist})$$

$$B_{ab} = \frac{1}{3} \Theta h_{ab} + \sigma_{ab} + \omega_{ab}$$

$$\begin{aligned}
* \xi^a \sigma_{ab} &= \xi^a B_{(ab)} - \frac{1}{3} \Theta \xi^a h_{ab} \\
&= 0 - \frac{1}{3} \Theta (\xi_{|a} \xi_{|b} - \xi_a \xi_b) = 0
\end{aligned}$$

$$\text{and } \xi^a \omega_{ab} = 0$$

$\leadsto$

$$\xi^b \nabla_b n^a = B^a_b n^b$$

$$\hookrightarrow B^a_b = \frac{1}{3} \Theta h^a_b + g^{ac} \sigma_{cb} + g^{ac} \omega_{cb}$$

$$\begin{aligned} \bullet g^{ac} \sigma_{cb} &= g^{ac} B_{(cb)} - \frac{1}{3} \Theta h^a_b \\ &= g^{ac} \frac{1}{2} (B_{cb} + B_{bc}) - \frac{1}{3} \Theta h^a_b \\ &= g^{ac} \frac{1}{2} (\nabla_b \xi_c + \nabla_c \xi_b) - \frac{1}{3} \Theta h^a_b \end{aligned}$$

\* Supposing that,  $\sigma = \omega = 0$  (for now)

$$\begin{aligned} \xi^b \nabla_b \eta^a &= \frac{1}{3} \Theta h^a_b \eta^b \\ &= \frac{1}{3} \Theta \eta^a \end{aligned}$$

$$\left( \xi^b \nabla_b - \frac{1}{3} \Theta \right) \eta^a = 0$$

$$\frac{d\eta^a}{d\tau} = \frac{1}{3} \Theta \eta^a$$

$$\ln|\eta| = \int \frac{1}{3} \Theta(\tau) d\tau \Rightarrow \eta^a = \hat{\eta}^a e^{\frac{1}{3} \int \Theta(\tau) d\tau}$$

and so  $\eta^a$  will vary based on  $\Theta(\tau)$  and so

- $\Theta$  measures the average expansion of v.l. nearby surrounding geodesics

Expansion:  $\Theta = \nabla_a \xi^a$

- Note  $\nabla_a \xi_b$  is normal to  $T_{\gamma(p)} M$   
define an infinitesimal ball in  
3D rest space orthogonal to the  
tangent  $\xi^a$

\* As defined in lec:  $h_{\mu\nu} = g_{\mu\nu} + l_\mu l_\nu$

$$g_{\mu\nu} = m_\mu m_\nu - l_\mu n_\nu$$

Claim:  $\Theta = \delta^{\alpha\beta} m_\alpha^\mu m_\beta^\nu \nabla_\mu l_\nu$

"proof"

$$\Theta = \nabla_\mu l_\nu h^{\mu\nu} = \nabla_\mu l_\nu (l^\mu l^\nu - l^{(\mu} n^{\nu)} + m^{(\mu} \bar{m}^{\nu)})$$

$$- \nabla_\mu l_\nu \frac{1}{2} (l^\mu n^\nu + l^\nu n^\mu) = 0$$

$$\text{from } l^2 = -1 \text{ and } l^a \nabla_a l^b = 0$$

$$\frac{1}{2} \nabla_\mu l_\nu (m^\mu \bar{m}^\nu + m^\nu \bar{m}^\mu)$$

$$\frac{1}{2} \nabla_\mu l_\nu (m_1^\mu + i m_2^\mu) (m_1^\nu - i m_2^\nu) + \dots$$

$$\frac{1}{2} \nabla_\mu l_\nu (m_1^\mu m_1^\nu + i (m_2^\mu m_1^\nu - m_1^\mu m_2^\nu) + m_2^\mu m_2^\nu)$$

$$+ \frac{1}{2} \nabla_\mu l_\nu (m_1^\mu m_1^\nu + i (m_2^\nu m_1^\mu - m_1^\nu m_2^\mu) + m_2^\nu m_2^\mu)$$

$\Rightarrow$  mixed terms cancel

$$\Rightarrow \Theta = \nabla_\mu l_\nu h^{\mu\nu}$$

$$L = \delta^{\alpha\beta} m_\alpha^\mu m_\beta^\nu \nabla_\mu l_\nu \quad \text{as claimed.}$$

Claim:  $\sigma_{\alpha\beta} = m_{[\alpha}^\mu m_{\beta]}^\nu \nabla_\mu l_\nu$  : Shear occurs in 2D subspace  
 "proof":

$$\begin{aligned} \sigma_{\mu\nu} &= \nabla_{(\mu} l_{\nu)} - \frac{1}{3} \theta h_{\mu\nu} \\ &= \nabla_{(\mu} l_{\nu)} - \frac{1}{3} \theta (-l_{(\mu} n_{\nu)} + m_{(\mu} \bar{m}_{\nu)} + l_{\mu} l_{\nu)}) \\ &= \nabla_{(\mu} l_{\nu)} - \frac{1}{3} \delta^{\alpha\beta} m_\alpha^\mu m_\beta^\nu \nabla_{\alpha\beta} l_\nu (-l_{(\mu} n_{\nu)} + m_{(\mu} \bar{m}_{\nu)} + l_{\mu} l_{\nu)}) \\ &= \nabla_{\alpha\beta} l_\nu \left( \delta_{\mu\alpha} \delta_{\nu\beta} - \frac{1}{3} \delta^{\alpha\beta} m_\alpha^\mu m_\beta^\nu (-l_{(\mu} n_{\nu)} + m_{(\mu} \bar{m}_{\nu)} + l_{\mu} l_{\nu)}) \right) \end{aligned}$$

$\Rightarrow$  Not too sure here, they go from  $(\mu\nu)$  coords

to only  $\alpha\beta$

Claim:  $\omega_{\alpha\beta} = m_{[\alpha}^\mu m_{\beta]}^\nu \nabla_\mu l_\nu$

$\rightarrow$  I think  $m_\alpha^\mu$  are used to achieve the above?

$$\sigma_{\alpha\beta} = m_\alpha^\mu m_\beta^\nu \sigma_{\mu\nu} ?$$

$$\begin{aligned} m_\alpha^\mu m_\beta^\nu \nabla_\mu l_\nu - \frac{1}{3} \theta m_\alpha^\mu m_\beta^\nu h_{\mu\nu} \\ = m_{[\alpha}^\mu m_{\beta]}^\nu \nabla_\mu l_\nu - \frac{1}{3} \theta \delta_{\alpha\beta} \end{aligned}$$

• Close, the trace term is still present.

\* All 3 terms are invariants of the null geodesic congruences.

\* Conditions for Petrov coeffs:

$$\begin{aligned} C_{\mu\nu\rho\sigma} l^\mu m^\nu \bar{l}^\rho \bar{m}^\sigma &= C_{\mu\nu\rho\sigma} l^\mu l^\rho (m^\nu \bar{m}^\sigma + l^\nu n^\sigma) \\ &= C_{\mu\nu\rho\sigma} l^\mu l^\rho g^{\nu\sigma} = 0 \end{aligned}$$

$$* \omega^{(1)} = \omega_\mu dx^\mu : \quad d^2 \omega = 0$$

$$0 = 2 \nabla_{[a} \nabla_{b} \omega_{c]} = R_{[abc]d} \omega_d$$

for  $\omega = \xi$  we can obtain equations for rate of change of  $(\theta, \sigma, \omega)$

$\Rightarrow$  From scratch

$$\frac{d}{d\tau} B_{ab} = \xi^c \nabla_c B_{ab} = \xi^c \nabla_c \nabla_b \xi_a$$

$$\hookrightarrow \nabla_c \nabla_b = \nabla_b \nabla_c + [\nabla_c, \nabla_b]$$

$$\begin{aligned} \frac{d}{d\tau} B_{ab} &= \xi^c \nabla_b \nabla_c \xi_a + \xi^c [\nabla_c, \nabla_b] \xi_a \\ &= \xi^c \nabla_b \nabla_c \xi_a + \xi^c R_{cbda} \xi_a \end{aligned}$$

$$= g \nabla_b \nabla_c \xi_a + \xi^c R_{cba} \xi_d$$

$$= \nabla_b (\xi^c \nabla_c \xi_a) - \underbrace{\nabla_b \xi^c \nabla_c \xi_a}_{B_b^c B_{ac}} + \xi^c R_{cba} \xi_d$$

$$\frac{d}{d\tau} B_{ab} - B_b^c B_{ac} + \xi^c R_{abc} \xi_d$$

Taking the Trace:

$$g^{ab} \frac{d}{d\tau} B_{ab} = -B^{ac} B_{ac} + g^{ab} \xi^c R_{abc} \xi_d$$

$$= -B^{ac} B_{ac} - R_{ab} \xi^a \xi^b$$

• LHS:  $\xi^c \nabla_c \nabla_b \xi_a h^{ab} = \xi^c \nabla_c \theta$

$$\xi^c \nabla_c \theta = \frac{d\theta}{d\tau}$$

RHS:  $-B^{ac} B_{ac} = -\left(\frac{1}{3} \theta h^{ab} + \sigma^{ab} + \omega^{ab}\right) \left(\frac{1}{3} \theta h_{ab} + \sigma_{ab} + \omega_{ab}\right)$

$$= -\left(\frac{1}{3} h^2 \theta^2 + \sigma^2 + \omega^2\right)$$

\* Here  $h^2 = 3$  (Gauged fixed to remove 2 dof from  $\xi \xi$  term)

$$\begin{pmatrix} g^{ab} + \xi^a \xi^b \\ 4 + \xi^2 + \xi^2 + 1 \\ 4 - 1 - 1 \end{pmatrix} \begin{pmatrix} g_{ab} + \xi_a \xi_b \\ \\ \end{pmatrix} \rightarrow 3$$

•  $-\frac{1}{3} \theta^2 - \sigma^2 + \omega^2$

$$B_{[ab]} g^{ac} g^{bd} B_{[cd]}$$

$$= \frac{1}{2} (B_{ab} - B_{ba}) g^{ac} g^{bd} B_{[cd]}$$

$$= \frac{1}{4} (B_{ab} - B_{ba}) (B_{cd} - B_{dc}) g^{ac} g^{bd}$$

$$= \frac{1}{4} (B_{ab} B_{cd} - B_{ba} B_{cd} - B_{ab} B_{dc} + B_{ba} B_{dc}) g^{ac} g^{bd}$$

$$= \frac{1}{4} (B^{cd} B_{cd} - B^{dc} B_{cd} - B^{cd} B_{dc} + B^{dc} B_{dc})$$

$$= \frac{1}{2} (B^{ab} B_{ab} - B^{ab} B_{ba})$$

\*  $B^{ab} B_{[ab]} = B^{ab} \omega_{ab} + B^{ba} \omega_{ab} - B^{ba} \omega_{ab}$

$$= 2 \omega^{ab} \omega_{ab} + B^{ba} \omega_{ab}$$

$$B^{ab} \omega_{ab} = 2 \omega^{ab} \omega_{ab} - B^{ab} \omega_{ab}$$

$\therefore B^{ab} \omega_{ab} = \omega^{ab} \omega_{ab}$

\* Note  $B^{ab} B_{ba} \rightarrow \omega^{ab} \omega_{ba} = -\omega^{ab} \omega_{ab}$

so we get a  $\textcircled{F}$

$$\xi^a \nabla_a \theta = \frac{d\theta}{d\tau} = -\frac{1}{3}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{ab}\xi^a\xi^b$$

## (Raychaudhuri's Equation)

• Positivity property of  $R_{ab}\xi^a\xi^b$

→ Use Einstein's Equations

$$R_{ab} - \frac{1}{2}g_{ab}R = 8\pi T_{ab}$$

$$R - 2R = 8\pi T \Rightarrow R = -8\pi T$$

$$R_{ab} + 4\pi g_{ab}T = 8\pi T_{ab}$$

$$R_{ab} = 8\pi \left[ T_{ab} - \frac{1}{2}g_{ab}T \right]$$

$$R_{ab}\xi^a\xi^b = 8\pi \left[ T_{ab} - \frac{1}{2}g_{ab}T \right] \xi^a\xi^b$$

$$= 8\pi \left[ \xi^a T_{ab} \xi^b + \frac{1}{2}T \right]$$

$T^{\mu\nu}$

$\mu$ : what is flowing

$\nu$ : through which direction/surface



$T^{00}$ : energy in time  $\sim \rho c^2$

$T^{0i}$ : Energy in space

•  $\xi^a$  points along  $\tau$  so  $\xi^a T_{ab} \xi^b$  is  $\sim T^{00}$

$\xi^a T_{ab} \xi^b$ : (Energy/vol) of matter as measured by an observer with 4-vel.  $\xi^a$

• Classical Matter:  $T_{ab}\xi^a\xi^b \geq 0$  (Weak energy Condition)

for all timelike  $\xi^a$

$\xi^a T_{ab} \xi^b + \frac{1}{2}T \geq 0$  (Strong energy Condition)

→ Unit timelike  $\xi^a$

\* 2 sphere in stereographic coords (ASIDE)

$$d\Sigma^2 = \frac{4dzd\bar{z}}{(1-|z|^2)^2}$$

$$d\Sigma^2 = d\theta^2 + \sin^2(\theta) d\phi^2$$

$$z = e^{i\phi} \tan(\frac{\theta}{2}), \quad x = \sin(\theta) \cos(\phi), \quad y = \sin(\theta) \sin(\phi)$$

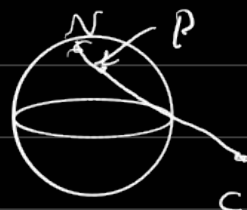
Stereographic projection  $z$

\*  $N: (0, 0, 1)$

$P: (\bar{X}, \bar{Y}, \bar{Z})$

$C: (x, y, 0)$

$(0, 0, 1) + t(x, y, z-1)$



$\hookrightarrow z=1$  for  $z=1$  case  $\rightarrow$

$(tx, ty, 1+t(z-1)) = \ell(t)$

at " $z=0$ ":  $\frac{1}{1+t(z-1)}$

$1+t(z-1) = 0 \quad t = 1-z$

$\begin{cases} x = t\bar{X} = \frac{1}{1-z}\bar{X} \\ y = t\bar{Y} = \frac{1}{1-z}\bar{Y} \end{cases}$

$\begin{cases} \bar{X} = x(1-z) \\ \bar{Y} = y(1-z) \end{cases}$

$\bar{X} = x(1-z)$

$\bar{Y} = y(1-z)$

$(x^2 + y^2)(1-z)^2 + z^2 = 1$

$r^2(1-z)^2 + z^2 = 1$

$z^2(1-2z+z^2) + z^2 = 1$

$z^2(1+z^2) - 2z^2z + (1^2 - 1) = 0$

$z = \frac{z^2 - 1}{(z^2 + 1)}$

$1-z = \frac{2}{1+z^2}$

$\frac{\bar{X}}{1-z} = \frac{2x}{1+z^2}$

$\frac{\bar{Y}}{1-z} = \frac{2y}{1+z^2}$

$z = \frac{1+z^2-1}{1+z^2}$

$\frac{4}{(1+z^2)^2} (dx^2 + dy^2)$

$ds^2 = \frac{4}{(1+z^2)^2}$

$U = \pi/2$

Const.  $V$

Const.  $U$

$i^+, i^-, i^0$ : Points

$z^+, z^-$ : 3D Mflds

increasing  $U$

$U = -\pi/2$

$U = \pi/2$

increasing  $V$

$U = -\pi/2$

\* The Boundary is not a mfd but its constituents are

$d\tilde{s}^2|_0 = 0$

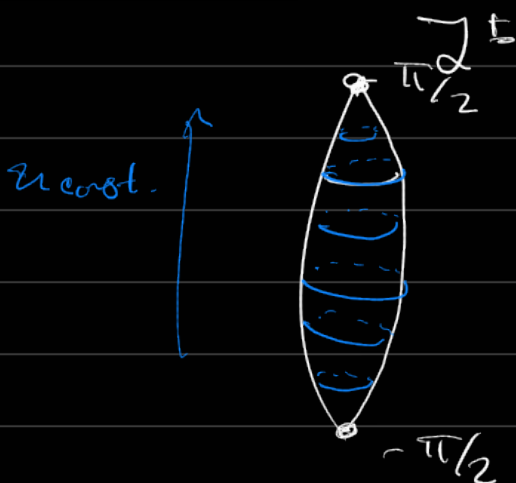
$4dz - d\bar{z}$

$$ds^2|_{\gamma^-} = \alpha (dU)^2 + \alpha \cos^2(U) (1 + |z|^2)^2$$

$$U \rightarrow \pm \pi/2$$

$$dS_{(2)}^2$$

Radius of  $S^2 \rightarrow 0$  as  $U \rightarrow \pm \frac{\pi}{2}$



$$N_{\mathbb{R}^+} \tilde{\mathcal{M}} : n = g^{\mu\nu} \partial_\nu \Omega \frac{\partial}{\partial x^\mu} \Big|_{\mathbb{R}^+}$$

$$= \alpha \cos(U) \partial U$$

only defined up to the  $r=0$  radius of the sphere

$$x^a = (U, z, \bar{z})$$

$$(n^a \partial / \partial x^a, h_{ab} dx^a dx^b) \sim (f^{-1} n^a, f^2 h)$$

~~Universal structure of  $\mathcal{I}_u$~~   
~~Carrollian~~

Note:  $h_{\alpha\beta} = f^2 h_{\alpha\beta}$

$$\Rightarrow \tilde{n}^a \Big|_{\mathbb{R}^+} = f^{-2} g^{\mu\nu} \tilde{\nabla}_\mu (f \Omega)$$

$$= f^{-2} g^{\mu\nu} (\tilde{\nabla}_{\mu\nu} f) \Omega + f^{-1} n^a$$

first term vanishes from  $\Omega|_{\mathbb{R}^+} = 0$

\* Gauge freedom from  $[n^a, h]$  allows us to choose  $f$  to remove factors in E.C.M (note)

BMS Coordinates:  $ds^2 = -dt^2 + dr^2 + r^2 d\Sigma^2$

$$u = t - r, \quad \Omega = r^{-1}, \text{ keep } (\sigma, \vartheta, \phi)$$

(Bondi-coords)

$$ds^2 = -du^2 + 2du dr + r^2 d\Sigma^2$$

$$\bullet du = dt - dr$$

$$-dt^2 = -du^2 + 2dr du = dr^2 \quad \checkmark$$

Factor out  $r^2$

$$ds^2 = -dU^2 + 2dU dr + r^2 d\Sigma^2$$

$$\Rightarrow (U, \Omega, \theta, \phi) \quad -\Omega^{-2} d\Omega = dr$$

$$d\sigma^2 = -dU^2 = 2d\tau d\Omega \Omega^{-2} d\Omega + \Omega^{-2} d\Sigma^2$$

$$= \Omega^{-2} (-\Omega^2 - 2d\tau d\Omega + d\Sigma^2)$$

$$d\tilde{\sigma}^2 = \Omega^2 d\sigma^2$$

### Continuation of Chap 9:

Weyl tensor: Trace free part of  $R_{\mu\nu\rho\sigma}$

$$C_{abcd} = R_{abcd} - \frac{2}{n-2} (g_{ac} R_{db} - g_{bc} R_{da}) + \frac{2R}{(n-1)(n-2)} g_{ac} g_{db}$$

Measures the tidal and conformal distortion of S.T. that is not locally determined by matter.

$$R^a{}_{acd}, R^a{}_{bad}, R^a{}_{bca}, \leadsto R_{ab} \text{ etc by } =$$

$$R_{ab}{}^c{}_c, R_{cb}{}^d{}_d, R_{ac}{}^d{}_d$$

$$(c) R^a{}_a = 0, R_{ab}{}^c{}_c = 0$$

left with

$$R^a{}_{bad}, R^a{}_{bca}, R_{cd}{}^c{}_d, R_{ec}{}^c{}_d$$

$$(i) g^{ac} R_{abcd}$$

$$(ii) g^{ad} R_{abcd}$$

$$(iii) g^{bc} R_{abcd}$$

$$(iv) g^{bd} R_{abcd}$$

$\leadsto \frac{1}{n-2}$  must be the combinatoric symmetry factor (2 cases from [1.3]).

(b) again from above we have (i)  $\rightarrow$  (iv) to obtain  $R$ . The symm factor  $\frac{1}{n-2}$  stays but is also now mult. by  $\frac{1}{n-1}$

$$\frac{d}{d\tau} B_{ab} = -B^c{}_b B_{ac} + R_{cba}{}^d \xi^c \xi_d$$

$$\hookrightarrow \text{Symmetric: } \sigma_{ab} = B_{(ab)}, \quad -\frac{1}{3} \Theta h_{ab} \quad \omega_{ab} = B_{[ab]}$$

$$\frac{d}{d\tau} B_{(ab)} = \xi^c \nabla_a B_{(cb)}$$

$$\frac{d}{d\tau} \left[ \sigma_{ab} + \frac{1}{3} \Theta h_{ab} \right] : \text{LHS}$$

RHS:

$$-B^c{}_{(b} B_{a)c} + R_{c(ba)}{}^d \xi^c \xi_d$$

$$= -g^{cd} B_{d(b} B_{a)c} + R_{c(ba)}{}^d \xi^c \xi_d$$

$$= -g^{cd} \left[ \left( \frac{1}{3} \Theta h_{db} + \sigma_{db} + \omega_{db} \right) \left( \frac{1}{3} \Theta h_{ac} + \sigma_{ac} + \omega_{ac} \right) \right. \\ \left. + \quad b \leftrightarrow a \right]$$

$$= -\frac{g^{cd}}{2} \left[ \frac{1}{3} \Theta^2 h_{db} h_{ac} + \frac{1}{3} \Theta (\sigma_{db} h_{ac} + \sigma_{ac} h_{db}) + \omega_{db} \omega_{ac} + \omega_{ac} \omega_{db} \right]$$

$$+ \sigma_{ab} \frac{1}{3} \theta h_{ac} + \sigma_{ab} \sigma_{ac} + \sigma_{ab} \omega_{ac} \\ + \omega_{ab} \frac{1}{3} \theta h_{ac} + \omega_{ab} \sigma_{ac} + \omega_{ab} \omega_{ac}]$$

Sum should be over c

$$\begin{aligned} & \bullet (g_{db} + \xi_d \xi_b) g^{cd} (g_{ac} + \xi_a \xi_c) \\ & \sim (\delta_b^a + \xi^a \xi_b) (g_{ac} + \xi_a \xi_c) \\ & = g_{bc} + \xi_c \xi_b + \xi_b \xi_c - \xi_b \xi_c \\ & = h_{bc} \end{aligned}$$

$$\bullet \frac{1}{3} \theta g^{ad} h_{db} \sigma_{ac} = \frac{1}{3} \theta (\delta_b^a \sigma_{ac}) = \frac{1}{3} \theta \sigma_{ba} \quad \text{blue}$$

$$\bullet \frac{1}{3} \theta g^{ad} h_{db} \omega_{ac} = \frac{1}{3} \theta \delta_b^a \omega_{ac} = \frac{1}{3} \theta \omega_{ba} \quad \text{green}$$

$$\bullet \frac{1}{3} \theta \sigma_{ab} \quad \text{blue}$$

$$\bullet g^{ad} \sigma_{db} \sigma_{ca}$$

$$\bullet g^{cd} \sigma_{db} \omega_{ac}$$

$$\bullet \frac{1}{3} \theta \omega_{ab} \quad \text{green}$$

$$\bullet g^{ad} \omega_{db} \sigma_{ca}$$

$$\bullet g^{ad} \omega_{db} \omega_{ca}$$

$$= \left( \frac{1}{4} \theta^2 h_{ba} + \frac{2}{3} \theta \sigma_{ba} + g^{ad} \sigma_{db} \sigma_{ac} + \dots \right. \\ \left. + g^{cd} \sigma_{db} \omega_{ac} + g^{ad} \omega_{db} \sigma_{ca} \right)$$

• Since  $g_{ab}$  is not positive definite, the map  $T_a^a$  is not diagonalizable.

• The eigenvectors of  $T_a^a$  with different eigenvalues are orthogonal.

→ In the diagonal case  $T_{ab}$

$$T_{ab} = \rho t_a t_b + p_1 x_a x_b + p_2 y_a y_b + p_3 z_a z_b \\ \{t^a, x^a, y^a, z^a\}.$$

$\rho$ : Rest energy density of matter

$p_i$ : principal pressures

\*  $T$  to be decomposable into its spectra

$$T = \lambda_i e_i \otimes e_i \quad T: \text{Self-Adjoint.}$$

$$\bullet T = g^{ab} T_{ab} = -\rho^2 + (x^2) p_1 + p_2 y^2 + p_3 z^2$$

• Weak energy condition:

$$T_{ab} \xi^a \xi^b \geq 0$$

$$(\xi^t)^2 \rho + P_1 (X \xi)^2 + P_2 (Y \xi)^2 + P_3 (Z \xi)^2 \geq 0$$

$$(i) \quad \rho \geq 0$$

$$(ii) \quad \rho + P_i \geq 0$$

• In the observer  $\xi^\mu = \gamma(1, v_x, v_y, v_z)$

$$\xi^\mu \xi_\mu = \gamma^2(-1 + v^2) = -\gamma^2(1 - v^2) = -1$$

(a) at rest:  $v_i = 0 \Rightarrow \rho \geq 0$

(b) Approaching speed of light along one axis

$$v_x = v_y = 0$$

$$\xi^\mu = \gamma(1, 0, 0, v_z)$$

$$\xi^\mu T_{\mu\nu} \xi^\nu = T_{00}(\xi^0)^2 + T_{33}(\xi^3)^2 \\ = \rho \gamma^2 + P_3 \gamma^2 v_z^2 \geq 0$$

$$\therefore \rho + P_3 v_z^2 \geq 0$$

$$\lim_{v_z \rightarrow 1} 0 \leq \rho + P_3 v_z^2 \leq \rho + P_3$$

$$\Rightarrow \rho + P_i \geq 0$$

Strong energy condition:  $T_{ab} \xi^a \xi^b + \frac{1}{2}T \geq 0$

$$\rho + \sum P_i \geq 0, \quad \rho + P_i \geq 0$$

• If Einstein equations hold, then strong energy conditions imply  $R_{ab} \xi^a \xi^b \geq 2 \left( T_{ab} \xi^a \xi^b + \frac{1}{2}T \right) \geq 0$

$$\text{and so } -R_{ab} \xi^a \xi^b \leq 0$$

• Interpreted as a manifestation of the attractiveness of gravity

• If the congruence is hypersurface orthogonal then  $B_{[ab]} = 0$

•  $-\sigma_{ab} \sigma^{ab}$ : manifestly non-positive

$$\frac{d\theta}{dt} + \frac{1}{3}\theta^2 = - \left( \sigma_{ab} \sigma^{ab} + R_{ab} \xi^a \xi^b \right) \leq 0$$

$$\hookrightarrow \theta \neq 0 \Rightarrow \theta^{-2} \frac{d\theta}{dt} + \frac{1}{3} \leq 0$$

$$\text{or } -\frac{d(\theta^{-1})}{dt} + \frac{1}{3} \leq 0$$

$$\frac{d(\theta^{-1})}{dt} \geq \frac{1}{3}$$

$$\theta^{-1} - 1 \geq (\theta^{-1}) - 1$$

$$\Theta(\tau) = (\Theta_0)^{-1} + 3\tau$$

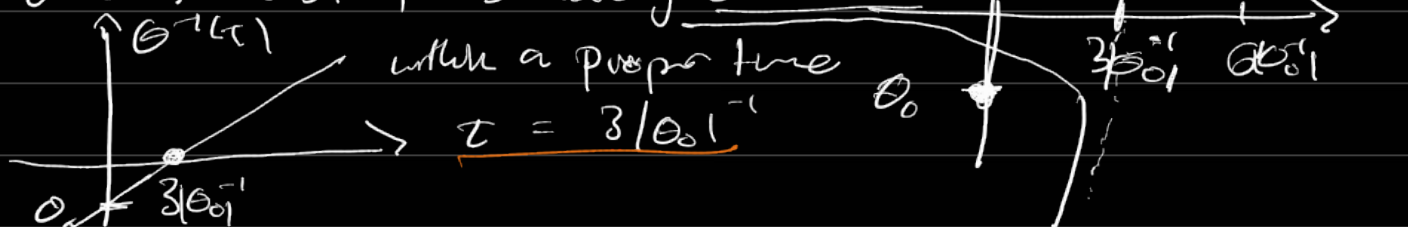
$$\Theta_0 = \Theta(0)$$

- Suppose  $\Theta_0 < 0$ : This means the congruence is initially convergent

$$\Theta(\tau) \leq \frac{1}{\Theta_0^{-1} + \frac{1}{3}\tau}$$

$$\Theta \xrightarrow{\tau \rightarrow \pm\infty} 0$$

$\Theta'(\tau)$  must pass through zero



Lemma: Let  $\xi^a$  be the tangent field of a hypersurface orthogonal timelike geodesic congruence ( $u_{[a}u_{b]} = 0$ ). Suppose (WEC)  $R_{ab}\xi^a\xi^b \geq 0$ , as will be the case if Einstein's eq hold in a spacetime where SEC ( $T_{ab}\xi^a\xi^b + \frac{1}{2}T \geq 0$ ) is satisfied by matter. If the expansion  $\Theta$  takes the negative value  $\Theta_0$  at any point on a geodesic in the congruence, then  $\Theta$  goes to  $-\infty$  along that geodesic within a proper time  $\tau \leq 3/|\Theta_0|$ .

$\Rightarrow$  Singularity in the congruence.

Null Geodesics:

In the timelike case:  $n^a \perp \xi^a$

$$\xi_a n^a = 0$$

$$(1) \frac{d}{d\tau}(\xi_b n^b) = \xi^a \xi_b \nabla_a n^b$$

$$(\nabla_{\xi} n)^b = [\xi, n]^b = \xi^a \nabla_a n^b - n^a \nabla_a \xi^b$$

$$\Rightarrow \xi_b (\nabla_{\xi} n)^b + n^a \nabla_a \xi^b$$

• If  $\xi^a \xi_a = \text{const} \Rightarrow$  second term vanishes

$$\therefore \frac{d}{d\tau}(\xi_b n^b) = 0$$

so  $\xi_b n^b$  is constant along each geodesic

(2) Deviation vectors which differ only by a multiple of  $\xi^a$  represent the same nearby geodesic

\* For  $N^a$ :  $K^a K_a = 0$

$$(i) \frac{d}{d\lambda}(K_b N^b) = K^a \nabla_a (K_b N^b) = K_b \nabla_a N^b + N^a K_b \nabla_a K^b$$

so  $K^a N_a$  does not vary along each geodesic

$$\bullet N^a = \hat{A}^a + \hat{B}^a$$

$$\Rightarrow \hat{A}^a: \hat{A}^a K_a \neq 0 \quad \text{s.t.} \quad \frac{d\hat{A}}{d\lambda} = \alpha \hat{A} \quad \text{along } K$$

$$\bullet K^a \nabla_a \hat{A}^b = \alpha \hat{A}^b$$

$$\bullet \hat{B}^a: \hat{B}^a K_a = 0 \quad \text{unique}$$

$\Rightarrow$  There is no natural way to do this decomposition

• "non-orthogonal" part of  $N^a$  ( $\hat{A}$ ) is uninteresting

since it remains similar along the geodesic

\* Only consider  $\hat{B}^a K_a = 0 \Leftrightarrow N^a K_a = 0$

(ii) Deviation vectors which differ only by a multiple of  $K^a$  ( $\tilde{N}^a = N^a + \alpha K^a$ ) represent a displacement of nearby geos.

$\Rightarrow$  Interesting quantity is  $[N^a]$  set.

$$\tilde{N}^a \sim N^a \Leftrightarrow \tilde{N}^a = N^a + \alpha K^a$$

so Reduces  $[N^a]$  to a 2D subspace

• The tangent vectors which are orthogonal to  $K^a$  (at  $T_p M$ ) form a 3D subspace  $\tilde{V}_p$

Let  $T_p M$ , then  $V^M \in T_p M$ . Let  $K^M \in T_p M$

$$\text{be null: } K^M K_M = 0$$

$$K^M = \left( \frac{dt}{d\lambda}, \frac{dx}{d\lambda}, \frac{dy}{d\lambda}, \frac{dz}{d\lambda} \right)$$

$$\bullet ds^2 = -dt^2 + (dx^i)^2 = 0$$

$$|V|^2 = 1$$

$$K^M = (1, V^i)$$

• Suppose  $K^M \sim (1, 0, 0, 1)$

$\hookrightarrow$  Set of vectors orthogonal to  $K^a$  are

$$V_x = (0, 1, 0, 0) \quad \text{and one other}$$

$$V_y = (0, 0, 1, 0)$$

•  $K^M$  is self orthogonal which is part of this

subspace:

$$\Rightarrow \hat{V}_p \equiv \text{span}\{V_x, V_y, K^a\}$$

\* let  $x^a, y^a \in \hat{V}_p$ , define  
 $x^a \sim y^a \Rightarrow \exists c \in \mathbb{R}$   
 $x^a - y^a = c K^a$

$\Rightarrow K^a$  is the zero element / vector  
 $\hat{V}_p \equiv \{[x^a] \mid x^a \in \hat{V}_p\}$

Claim,  $\dim(\hat{V}_p) = 2$

$\Rightarrow$  Let  $u \in \hat{V}_p$ , then

$$y \in [V_x] \Rightarrow y^a - V_x^a = c K^a$$

$$y = V_x^a + c K^a$$

$$x \in [V_x] \Rightarrow x^a = V_x^a + c K^a$$

$\Rightarrow [K^a]$ : Zero element,

$$u = \alpha V_x + \beta V_y + \gamma K^a$$

$$\Rightarrow [u] = \alpha [V_x] + \beta [V_y] \quad V_x = (0, 1, 0, 0)$$

$$\hat{V}_p \equiv \text{span}_{\mathbb{R}}\{[V_x], [V_y]\} \quad V_y = (0, 0, 1, 0)$$

Zero element comes from the field

\*  $n^a \in \hat{V}_p$  As a result of the above analysis  
 (In the case of 2-orientation)

Let-3:  $\{x^a\}$  coordinates on  $\mathcal{Z}$ ,  $\{x^m\}$  on  $\mathcal{M}$

$(\Omega, g)$ : defined up to an equivalence class

$$(\Omega, g) \sim (\lambda(x)\Omega, \lambda^2 \Omega^2 g)$$

$$(n, h|_Z) \sim (\lambda^{-2} n^a, \lambda^2 h)$$

• Proposition:  $n^m$  is null:  $n^m = \tilde{g}^{mn} \tilde{\nabla}_n \Omega$

$\Rightarrow \tilde{n}^m$  is both tangent and normal to  $\mathcal{Z}^+$ ...

$$\tilde{n}^m = \tilde{g}^{mn} \tilde{\nabla}_n \Omega = \Omega^{-2} g^{mn} \partial_n \Omega$$

so the lecture uses  $n^m = \Omega^2 \tilde{n}^m$

Thus,  $n^m$  is tangent to  $\mathcal{Z}^+$  since

$$n^m \nabla_n \Omega = g^{mn} \nabla_n \Omega \nabla_m \Omega = 0$$

proof:  $n n_{ab} = g^{\mu\nu} \nabla_\nu \Omega g_{\mu\sigma} \nabla^\sigma \Omega = 0$   
 [since  $\nabla_\mu \Omega|_J = 0$ ,  $n \equiv n^a \partial/\partial x^a$ ]

• "Induced metric" has  $dx^a dx^b$  is degenerate

$h_{ab} n^a = 0$  (Signature  $(0, +, +)$ )

$\hookrightarrow h_{ab} h^{ac} \nabla_c \Omega = \delta_b^c \nabla_c \Omega = \nabla_b \Omega|_J = 0$

•  $\mathcal{L}_n h_{ab} \propto h_{ab}$

$(\mathcal{L}_n h)_{ab} = \nabla_a n_b + \nabla_b n_a = \nabla_a \nabla_b \Omega + \nabla_b \nabla_a \Omega$   
 $= 2 \nabla_a \nabla_b \Omega$

• For spatial  $\omega$ , we require  $\nabla_a \nabla_b \Omega|_{i^0} = 2g_{ab}(i^0)$

$\hookrightarrow$  Also, from Wald we can gauge  $\nabla_a \nabla_b \Omega|_{J^+} = 0$   
 when means  $(\mathcal{L}_n h)_{ab} = 0$  but  $g \neq 0$  at  $J^+$

So im not too sure where the claim comes from

Back to Wald Chp 9:  $t^a \in T_p M$  does not naturally give rise to an element in  $\hat{V}_p$

• However, let  $t \in \hat{V}_p$ , then

$t^a K_a = 0$

$t^a = \alpha V_x^a + \beta V_y^a + \gamma K^a$

$\hat{V}_p \longrightarrow \hat{V}_p$

$t^a \longmapsto \{t^a\}$

• Dual vector  $\mu_a \in T_p^* M$  naturally gives rise to a dual  $\tilde{\mu}_a \in (\tilde{V}_p)^*$  by

and let  $\mu \in T_p^* M$

$\mu: T_p M \rightarrow \mathbb{R}$

$\mu(V) := \begin{cases} 0 & , V \notin \tilde{V}_p \sim \mu_N \\ \mu_N V^a, & V \in \tilde{V}_p \sim \tilde{\mu} \end{cases}$

\*  $\mu$  induces a normal vector to the complement of  $\tilde{V}_p$ :  $\mu = \mu_N \oplus \tilde{\mu}$

so  $\mu$  induces  $\tilde{\mu}$

\*  $\tilde{\mu}$  induces a dual vector on  $\hat{V}_p \ni \hat{\mu}$  iff

$\tilde{\mu}_a K^a = \mu_a K^a = 0.$

$$\tilde{\mu} \in \tilde{V}_p : \quad \tilde{\mu}_a t^a, \quad t^a k_a = 0$$

( $\Rightarrow$ ) Suppose  $\tilde{\mu}$  induces  $\hat{\mu}$ , then  $\hat{\mu}$  exists

$$\hat{\mu}([t]) := \hat{\mu}_a t^a$$

Since  $\hat{\mu} \in \hat{V}_p^*$   $g^{ab} : \hat{V}_p^* \rightarrow \hat{V}_p$

$$\hat{\mu}_a g^{ab} \in \hat{V}_p$$

$\Rightarrow$  on  $\hat{V}_p$  we define  $\hat{g} : \hat{V}_p \times \hat{V}_p \rightarrow \mathbb{R}$

$$\text{via } \hat{g}([\alpha], [\beta]) := g(\alpha, \beta)$$

Must show that this is representative indep.

$$[\alpha] := \{x \in \tilde{V}_p \mid x \sim \alpha\}$$

Let  $x \in [\alpha]$  and  $y \in [\beta]$

$$\begin{aligned} g(x, y) &= g(\alpha + c_1 \kappa, \beta + c_2 \kappa) \\ &= g(\alpha, \beta) + c_2 g(\alpha, \kappa) + c_1 g(\kappa, \beta) \\ &\quad + c_1 c_2 g(\kappa, \kappa) \end{aligned}$$

Since  $\alpha, \beta \in \tilde{V}_p$ , then  $g(\alpha, \kappa) = 0$  and so

$$\hat{g}([x], [y]) = g(x, y) = g(\alpha, \beta) = \hat{g}([\alpha], [\beta])$$

So  $\hat{g}$  is well defined.

Note then that all elements of  $\tilde{V}_p$  and  $\hat{V}_p$  are such that:

$$\tilde{\mu}_a \hat{g}^{ab} k_b = 0$$

but since  $\hat{\mu}$  is induced/restricted from  $\tilde{\mu}$  then  $\hat{\mu}^a = \tilde{\mu}^a - \alpha k^a$

$$\tilde{\mu}_a \hat{g}^{ab} k_b = 0 = (\tilde{\mu}^a - \alpha k^a) k_a = \tilde{\mu}^a k_a = 0$$

$$\therefore \text{Note } (\mu - \mu_N)^a k_a = 0$$

$$\text{or } \mu^a k_a = \mu_N^a k_a$$

Which vanishes only if  $\mu_N = 0$  i.e.  $\mu$  is an element of  $\tilde{V}_p$   $\mu = \tilde{\mu}$

• Might need to do a proof by contradiction here.

$$(\Leftarrow) \text{ Suppose } \mu_a k^a = \tilde{\mu}^a k_a = 0$$

$$\text{then } (\mu_a - \tilde{\mu}_a) k^a = 0$$

$$\text{and so } \mu_a \in \tilde{V}_p \text{ and } \mu_N^a k_a = 0$$

Which will only occur if  $\mu = \tilde{\mu}$ .  $\square$

A tensor  $T_{a_1 \dots a_n b_1 \dots b_n}$  over  $\mathbb{R}^n$  naturally gives

rise to a tensor  $\hat{T} \in \hat{V}_p$  iff

$$T_{a_1 \dots a_n b_1 \dots b_n} K^c = T_{a_1 \dots a_n b_1 \dots b_n} K^c \equiv 0$$

Contracted with  $\hat{V}_p$  &  $\hat{V}_p^*$  elements

↳ This alone is a stronger statement.

→ For  $g_{ab}$ : Take  $V \in \hat{V}_p$  then

$$K^a V_a = 0 \text{ (trivially)}$$

also for  $V^a \in \hat{V}_p$ . So  $g$  naturally

\* Here  $\hat{h} \equiv \hat{g}$ , Note  $\hat{h}$  is acting on a 2D

subspace.

$$\text{note } \hat{h}([V_x] + [V_y], [V_x] + [V_y])$$

$$= h_{xx} + 2h_{xy} + h_{yy}$$

$$h_{xx} = 1, \quad h_{yy} = 1, \quad h_{xy} = 0?$$

$$\hat{h}_{ab} = \delta_{ab} \Rightarrow h = \hat{h}_{ab} dx^a dx^b$$

\* Take a null congruence generated by flows  $\gamma_s(t)$  which span  $\Sigma$  with coordinates  $(t, s)$ .

$$\boxed{B_{ab} \equiv \nabla_b K_a}$$

$$\left( \frac{d}{dt} + v_s(t) \right) \rightarrow s$$

Also satisfies the above property:

$$\textcircled{a} \quad V^a B_{ab} K^b = V^a \nabla_b K_a K^b = V^a (K^b \nabla_b K_a) = V^a \frac{d}{dt} K_a = 0 \quad \text{since } K_a = 1 \text{ on } \Sigma$$

$$\textcircled{b} \quad K^a B_{ab} V^b = K^a \nabla_b K_a V^b = 0$$

$$K^a K_a = 0 \Rightarrow 2 \nabla_b K^a K_a = 0$$

$\Rightarrow B$  gives rise to a metric form  $\hat{B}$ .

Decompose  $\hat{B}$  into

$$\hat{B}_{ab} = \left( \frac{1}{2} \theta \right) \hat{h}_{ab} + \hat{\sigma}_{ab} + \hat{\omega}_{ab}$$

$$\theta \equiv \hat{h}^{ab} \hat{B}_{ab}$$

$$\hat{\sigma}_{ab} \equiv \hat{B}_{(ab)} - \frac{1}{2} \theta \hat{h}_{ab}$$

• For timelike curves  $\gamma^u$ :  $h_{ab}$  spans the spatial dimensions i.e. a 3D subspace with  $h^2 = 3$

• Here, for null geodesic congruences  $h = g|_{\Sigma^\perp}$

induces a 2D metric  $\hat{h}$  on the equivalence class of  $\text{span}\{[e_1], [e_2]\}$  which is also "spatial" when in the frame  $k^\mu \sim (1, 0, 0, 1)$  via  $\hat{h}$ .

$\Rightarrow$  We have that  $\hat{h}^{ab} \hat{h}_{ab} = \underline{2}$ ,  $a, b = x, y = "1, 2"$

In the above frame  $\hat{h}_{ab} = \delta_{ab}$ ,  $a, b$

$\Rightarrow$  Now, let  $B_{ab} = \nabla_b k_a$ ,  $k^a = \frac{dx^a}{d\lambda}$

$$\begin{aligned} \frac{d}{d\lambda} B_{ab} &= k^c \nabla_c B_{ab} = k^c \nabla_c \nabla_b k_a \\ &= k^c \nabla_b \nabla_c k_a + k^c R_{cba}{}^d k_d \\ &= \nabla_b (\underbrace{k^c \nabla_c k_a}_0) - \nabla_b k^c \nabla_c k_a + k^c R_{cba}{}^d k_d \end{aligned}$$

$$= -B^c{}_b B_{ac} + k^c R_{cba}{}^d k_d$$

$$\bullet \quad k^c \nabla_c B_{ab} = -B^c{}_b B_{ac} + k^c R_{cba}{}^d k_d$$

$$\hookrightarrow k^c \nabla_c B_{ab} + B^c{}_b B_{ac} = k^c R_{cba}{}^d k_d$$

When indices

$$k^c \nabla_c \hat{B}_{ab} + \hat{B}^c{}_b \hat{B}_{ac} = R_{cba}{}^d k_d k^c$$

$$(\text{Expansion}) \quad \frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \hat{\sigma}^2 + \hat{\omega}^2 - R_{cd} k^c k^d$$

$$(\text{Twist}) \quad k^c \nabla_c \hat{\sigma}_{ab} = -\theta \hat{\sigma}_{ab} + \hat{C}_{cbad} k^c k^d$$

$$(\text{Shear}) \quad k^c \nabla_c \hat{\omega}_{ab} = -\theta \hat{\omega}_{ab}$$

$\hookrightarrow \hat{C}_{cbad} k^c k^d = 0$  If  $\hat{C}_{cbad} k^c k^d x^a y^b = 0 \quad \forall x, y \in \hat{V}_p$  and is equivalent to

$$k_e [C_b]{}_c [a] k_f] k^c k^d = 0$$

\* Thus the condition that  $k^a$  is a principal null vector of the Weyl tensor.

$\rightarrow$  For a tetrad  $(l^\mu, n^\mu, m^\mu, \bar{m}^\mu)$  the above is equivalently

$$\psi_0 = 0$$

$\Rightarrow$  For a null vector  $k^\mu$  the above is equivalent to

• If  $\chi$  has multiplicity  $\geq 1$  then additional Weyl scales vanish

Type I:  $\psi_0 = 0$

II:  $\psi_0 = \psi_1 = 0$

III:  $\psi_0 = \psi_1 = \psi_2 = 0$

IV:  $\psi_0 = \psi_1 = \psi_2 = \psi_3 = 0$

V:  $\psi_0 = \dots = \psi_n = 0$

$\{\psi_0, \dots, \psi_n\}$ : again define the S-E invariants of the Weyl tensor  $C$

• For  $K^a \nabla_c \hat{\sigma}_{ab} = -\theta \hat{\sigma}_{ab} + C_{cad} K^c K^d$ , if  $\hat{\sigma}_{ab} = 0$   
 $\frac{d}{ds} \hat{\sigma}_{ab} = 0$  and so  $C_{cad} K^c K^d = 0$ , then

$K^a$  is a principal null vector

↳ For  $R_{ab} = 0$ , if  $\hat{\sigma}_{ab} = 0$ , then  $K^a$  must be repeatedly a princ. null vector

⇒ Goldberg-Sachs theorem

\* For the expansion:  $\theta$

$$\frac{d\theta}{ds} = -\frac{1}{2}\theta^2 - \hat{\sigma}^2 + \hat{\omega}^2 - R_{cd} K^c K^d$$

look at  $R_{ab}$ ,  $R_{ab} = T_{ab} - \frac{1}{2}Tg_{ab}$

$$R_{ab} K^a K^b = 8\pi(T_{ab} - \frac{1}{2}Tg_{ab}) K^a K^b = 8\pi K^a T_{ab} K^b$$

$$\Rightarrow T_{ab} K^a K^b \geq 0$$

• For a timelike  $\xi^a$  if the Strong energy condition holds

$$T_{ab} \xi^a \xi^b - \frac{1}{2}T\xi^a \xi_a \geq 0$$

then by continuity (i.e. the massless limit)

$$\xi^a \xi_b \rightarrow 0 \quad T_{ab} K^a K^b \text{ holds}$$

• If WEC holds then the same is true.

For dustlike  $T_{ab} \sim \rho t_a t_b + p_i x_i^2$

The Necessary and Sufficient condition for satisfying

$$T_{ab} K^a K^b \geq 0 \quad (\forall K^a K_a = 0)$$

$$\rho + p_i \geq 0 \quad (i = 1, 2, 3)$$

Lemma: Let  $K^a$  be tangent field of a hypersurface orthogonal null geodesic congruence. Suppose  $R_{ab} K^a K^b \geq 0$ . If  $\theta$  takes the negative value  $\theta_0$  at any point on a

geodesic in  $Y_s(t)$ , then  $\theta$  goes to  $\infty$  along that geodesic within affine length  $\lambda \leq 2/|\theta|$

- Hyper surface orthogonal means  $h_{ab} = 0$ ,

Def: Let  $\gamma(\lambda)$  be a geodesic with tangent  $V^m = \frac{dx^m}{d\lambda}$ , then a solution to the geodesic deviation equation ( $\mathcal{N}^a$ )

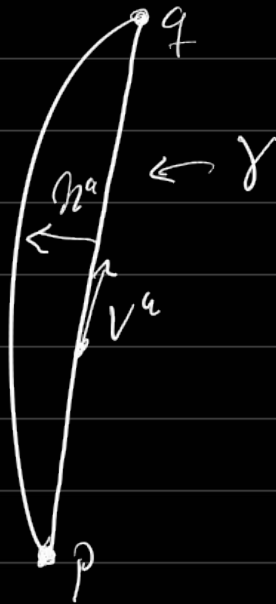
$$\frac{d^2 \mathcal{N}^c}{d\lambda^2} + R_{abcd} \mathcal{N}^b V^c V^d = 0$$

is a Jacobi-field on  $\gamma$ .

- A pair of points  $p, q \in Y$  are said to be conjugate if  $\exists$  a Jacobi field  $\mathcal{N}^a$ :

$$\mathcal{N}^a(p) = \mathcal{N}^a(q) = 0$$

and not identically zero elsewhere

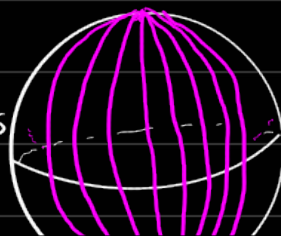


$\Rightarrow p, q$  are conjugate if an "inf. nearby" geodesic intersects  $\gamma$  at  $p$  and  $q$ .

ex: For  $S^2$ : Riemann Sphere  $N \in S$  are conjugate  $L$  of "longitudinal geodesics"

• The necessity of "another" geodesic is not a defect.

\* Characterize the stage at which a timelike geodesic fails to be a local maximum of



proper time between two points and a null geodesic fails to remain on the boundary of the future point.

\* Timelike geodesics:  $\gamma$ : timelike with  $\dot{\gamma}^a \dot{\gamma}_a = -1$   $\dot{\gamma}^a = \frac{dx^a}{d\tau}$   
 Let  $p \in \gamma$ . Consider congruence of all timelike geodesics passing through  $p$ . (Singular at  $p$ )

• Then  $R^a(p) = 0$

Claim: a point  $q \in \gamma$  lying in the future of  $p$  is conjugate to  $p$  iff the expansion  $\theta \rightarrow -\infty$  at  $q$ .

• Introduce an orthonormal basis of spatial vectors  $\{e_1^a, e_2^a, e_3^a\}$  orthogonal to  $\dot{\gamma}^a$  and parallelly propagated along  $\gamma$ .

$$\frac{d^2 R^a}{d\tau^2} = -R_{ABN}^a \dot{\gamma}^A R^B \dot{\gamma}^N$$

$R^a(\tau)$  will depend on

(a)  $R^a(0)$

(b)  $dR^a/d\tau(0)$

• For  $p=0$ ,  $R^a(0) = 0$

$\Rightarrow$

$$R^i(\tau) = -R_{\alpha\beta j}^i \dot{\gamma}^\alpha R^\beta \dot{\gamma}^j$$

$$R^i(\tau) = A_j^i(\tau) \frac{dR^j}{d\tau}(0)$$

$$\frac{d^2 A_j^i}{d\tau^2} = -R_{\alpha\beta\sigma}^i \dot{\gamma}^\alpha \dot{\gamma}^\sigma A_j^\sigma$$

$$\Rightarrow A_j^i(0) = 0 \text{ and } \frac{dA_j^i}{d\tau}(0) = \delta_j^i$$

$$\frac{dR^i(\tau)}{d\tau} \bigg|_{\tau=0} = \frac{dA_j^i(0)}{d\tau} \frac{dR^j(0)}{d\tau}$$

$$A_j^i = \tau \delta_j^i + O(\tau^2)$$

$$* R_{\alpha\beta\gamma\delta} = \frac{1}{2} (R_{\alpha\beta\gamma\delta} + R_{\alpha\gamma\beta\delta} + R_{\alpha\delta\beta\gamma} + R_{\alpha\gamma\delta\beta} + R_{\alpha\delta\gamma\beta} + R_{\alpha\beta\gamma\delta})$$

Claim:  $q$  is conjugate to  $p$  iff  $\exists \frac{dn^i}{d\tau}(0) \neq 0$  for  $L$  when  $N^i(q) = 0$  at  $q$ .

"proof": Occurs iff  $\det(A^i_j) = 0$  at  $q$

Let  $N^i(0) = 0$

$$N^i(\tau) = A^i_j \frac{dn^j}{d\tau}(0)$$

Let  $\tau_q$  be such that  $N^i(\tau_q) = N^i(q)$

$$0 = A^i_j(\tau_q) \frac{dn^j}{d\tau}(0)$$

If  $dn^i/d\tau = 0 \Rightarrow N^i = \text{constant}$  and  $N^i = 0$

$\forall \tau$  all all geodesics have already converged  
 $\Rightarrow$  Assume  $dn^i/d\tau(0) \neq 0$ , then

$$A^i_j(\tau_q) \frac{dn^j}{d\tau}(0) = 0$$

$$\Rightarrow \det(A^i_j) = 0$$

$$\frac{dn^i}{d\tau} A^i_j(\tau_q) \frac{dn^j}{d\tau} = 0$$

diagonalize  $\sim \det(A^i_j) = 0$

\*  $\det(A) = 0$  is the necessary and sufficient condition for a conjugate point to  $p$

• Between  $p, q$   $\det(A(\tilde{\tau})) \neq 0$   
 $\tilde{\tau} \in (\tau_p, \tau_q)$

$$\begin{aligned} \frac{dn^i}{d\tau} &= \xi^a \nabla_a N^i = \xi^a \nabla_a [(e_i)_b N^b] = (e_i)_b \xi^a \nabla_a N^b \\ &= (e_i)_b N^a \nabla_a \xi^b = (e_i)_b N^a B^b_a \\ &= N^a B^i_a \end{aligned}$$

$$\frac{dn^i}{d\tau} = \sum_{\alpha=1}^n B^i_\alpha N^\alpha \frac{dN^\alpha}{d\tau}(0)$$

$$\frac{dn^i}{d\tau} = \frac{dA^i_j}{d\tau} \frac{dn^j}{d\tau}(0) = B^i_\alpha N^\alpha(\tau)$$

$$\text{but } N^\alpha(\tau) = A^\alpha_j \frac{dn^j}{d\tau}(0)$$

$$\frac{dA^i_j}{d\tau} \frac{dn^j}{d\tau}(0) = B^i_\alpha A^\alpha_j \frac{dn^j}{d\tau}(0)$$

$$\therefore \frac{dA^i_j}{d\tau}(\tau) = B^i_\alpha A^\alpha_j(\tau) \Rightarrow \frac{dA}{d\tau} = BA$$

$$\frac{dA}{d\tau} = BA$$

$$B = \frac{d}{dt} A^{-1}$$

$$\Theta = \text{Tr}(B) = \text{Tr} \left[ \frac{dA}{dt} A^{-1} \right] =$$

$$\ln|\det(M)| = \text{Tr}(\ln(M))$$

$$\Theta = \frac{d}{dt} \text{Tr} [\ln(A)] = \frac{d}{dt} (\ln |\det(A)|)$$

$$\Theta = \frac{d}{dt} (\ln |\det(A)|)$$

with

$$n^i(t) = A^i_j(t) \frac{dx^j}{dt}(0)$$

$$\bullet \Theta \rightarrow -\infty \text{ at } q \text{ (i.e. } n^i(t_q) = 0)$$

$$\text{then } \frac{1}{\det(A)} \frac{d}{dt} (\det(A)) \rightarrow -\infty$$

$$\text{but } \frac{d}{dt} A \sim BA \text{ and } A \rightarrow 0 \text{ as } t \rightarrow t_q$$

$$\lim_{t \rightarrow t_q} (\det(A)) = 0$$

\* If  $\det(A) \rightarrow 0$  at  $q$ , it follows that  $\Theta \rightarrow -\infty$

• If a congruence of timelike geodesics passing through  $p$  is hypersurface orthogonal.

Hawking and Ellis (1973): Within a convex normal neighborhood of  $p$  the geodesics in this congruence are orthogonal to the surfaces of constant proper time  $\tau$  along the geodesics, and from  $\xi^c \nabla_c w_a = -\frac{2}{3} \Theta w_a - 2 \sigma^c_b w_a \xi_c$  it follows that if  $w_a$  vanishes at one time, it must vanish at all times

Proposition:  $(M, g)$  a spacetime satisfying Strong E.C. Condition  $R_{ab} \xi^a \xi^b \geq 0 \quad \forall \xi^a$ . Let  $V$  be a timelike geodesic and let  $p \in V$ . Suppose the congruence of the congruence of timelike geodesics emanating into the future from  $p$  attains a negative value  $\Theta_0$  at  $r \in V$ . Then within proper time

$\Gamma = S(100)$  from  $\Gamma$  along  $\Gamma \ni$  a point  $q$  conjugate to  $p$ , assuming that  $\gamma$  extends that far.

\* Fubini-Study Metric from field theory.

$\mathbb{CP}^n$ : As the embedding from  $X^M: \mathcal{M} \rightarrow \mathbb{C}^{n+1}$

\* Let  $X \in \mathbb{CP}^{n+1} \ni X'$

s.t.  $X^M = (X^1, \dots, X^{n+1})$  and  $X'^M = (X'^1, \dots, X'^{n+1})$

let  $\rho \in \mathbb{C}^*$

$$X \sim X' \Rightarrow X = \rho X'$$

\* If  $X^i \neq 0$  then we can define projective coordinates  $Z_{\{I\}}^{\alpha} = X^{\alpha} / X^I$ ,  $I = 1, \dots, n+1$

\* All lines with  $X^I \neq 0$ .

\* Take  $X: \mathcal{M}^D \rightarrow \mathbb{C}^{n+1}$

$$X(x) = (X^1, \dots, X^{n+1})$$

$$\bar{(X^I)}^* =: \bar{X}^I = ((X^1)^*, (X^2)^*, \dots, (X^{n+1})^*)$$

$$X^I \bar{X}_I = \alpha \in \mathbb{R}$$

$\hookrightarrow |\rho|^2 (X')^I (\bar{X}')_I = \alpha$  and so by choosing

$|\rho|^2 = \alpha$  then we can easily

impose the constraint  $X^I \bar{X}_I = 1$  without loss of generality. Meaning  $|\rho|^2 = 1$

\* This choice of  $\alpha$  is a choice of representative

\* The common phase of the fields  $X^I(x)$  is irrelevant by introducing a U(1) Gauge potential  $A_\mu(x)$

$$X^I \bar{X}_I = 1 \leadsto |e^{i\theta(x)} X^I e^{-i\theta} \bar{X}_I| = 1$$

$$\therefore \partial_\mu \rightarrow (\partial_\mu - iA_\mu)$$

\* Massless U(1) Complex Scalar:

$$\mathcal{L} = -(\partial_\mu X^I)(\partial^\mu X_I)^*$$

$$= -(\partial_\mu - iA_\mu)X^I (\partial^\mu + iA^\mu)\bar{X}_I$$

\* E.O.M for  $A_\mu$ :  $\delta \mathcal{L} = 0$

$$\frac{\delta}{\delta A_\mu} \left( -i A_\mu \bar{X}^\mu \partial^\mu X_I + i \partial_\mu \bar{X}^\mu A^\mu X_I - A_\mu A^\mu \bar{X}^\mu X_I \right) = 0$$

$$-i \bar{X}^\mu \partial^\mu X_I + i \partial_\mu \bar{X}^\mu X_I - 2 A^\mu \bar{X}^\mu X_I = 0$$

\* Subject to  $\bar{X}^\mu X_I = 1$  then

$$A^\mu = -\frac{i}{2} \left[ \bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I \right]$$

• Plugging this back into the Lagrangian:

$$\mathcal{L} = - \left[ \partial_\mu + i \left( -\frac{i}{2} \bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I \right) \bar{X}_I \right] \left[ \partial^\mu - i \left( -\frac{i}{2} \bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I \right) X_I \right]$$

$$= - \partial_\mu \bar{X}_I + \frac{1}{2} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I) \bar{X}_I \left( \partial^\mu X_I - \frac{1}{2} \bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I \right)$$

$$= - \left[ \partial_\mu \bar{X}_I \partial^\mu X_I + \frac{1}{2} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I) \bar{X}_I \partial^\mu X_I \right. \\ \left. - \frac{1}{2} \partial_\mu \bar{X}_I X_I (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I) - \frac{1}{4} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I)^2 \right]$$

$$= - \left[ \partial_\mu \bar{X}^\mu \partial^\mu X_I + \frac{1}{2} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I) \left[ \bar{X}_I \overleftrightarrow{\partial}^\mu X_I \right] - \frac{1}{4} \dots \right]$$

$$\mathcal{L} = - \partial_\mu \bar{X}^\mu \partial^\mu X_I - \frac{1}{4} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I)^2, \quad \bar{X}_I X_I = 1$$

\*  $\mathcal{L}$  as above is U(1) Gauge invariant under

$$X_I(x) \rightarrow X'^I(x) = e^{i\theta(x)} X_I(x)$$

$$\mathcal{L}' = -i \partial_\mu \theta \bar{X}^\mu \partial^\mu X_I - i \partial_\mu \theta \bar{X}^\mu \partial^\mu X_I$$

$$+ i \partial_\mu \bar{X}^\mu \partial^\mu \theta X_I - \partial_\mu \bar{X}^\mu \partial^\mu X_I$$

$$= -i \partial_\mu \theta \partial^\mu \theta - i \partial_\mu \theta \left[ \bar{X}^\mu \overleftrightarrow{\partial}^\mu X_I \right] - \partial_\mu \bar{X}^\mu \partial^\mu X_I$$

$$- \frac{1}{4} (\bar{X}^\mu \overleftrightarrow{\partial}_\mu X_I)^2$$

\*  $-\frac{1}{4} \dots$  Use  $X^\mu = \frac{1}{\sqrt{1+z^2}} (z^1, \dots, z^n, 1)$

$$z^\alpha \equiv X^\alpha / X^{n+1}, \quad \bar{z}^\alpha z_\alpha \equiv \bar{z}^\alpha \delta_{\alpha\beta} z_\beta$$

$\hookrightarrow \mathbb{CP}^n$  metric:

$$ds^2 = 2g_{\alpha\bar{\beta}} dz^\alpha d\bar{z}^\beta = 2 \left( \frac{1}{1+\bar{z}z} \right) \left( \delta_{\alpha\beta} - \frac{\bar{z}_\alpha z_\beta}{(1+\bar{z}z)} \right) dz^\alpha d\bar{z}^\beta$$

\* Given a Hermitian metric, fundamental 2-form:

$$\Omega = -2i g_{\alpha\bar{\beta}} dz^\alpha \wedge d\bar{z}^\beta \quad | \quad \Omega = \bar{\Omega}$$

(Kähler form)

Kähler mfd:  $d\Omega = 0$

$$d\Omega = -i(\partial_\gamma g_{\alpha\bar{\beta}} - \partial_\alpha g_{\gamma\bar{\beta}}) dz^\gamma \wedge dz^\alpha \wedge d\bar{z}^\beta + c.c.$$

(Kähler)  
condition

$$\partial_{\bar{x}} g_{\alpha\bar{\beta}} - \partial_{\alpha} g_{\bar{x}\bar{\beta}} = 0 \Rightarrow \partial_{\bar{x}} g_{\alpha\bar{\beta}} = 0 \quad (\text{Cauchy})$$

$\Rightarrow$  locally we can express  $g_{\alpha\bar{\beta}}$  as

$$g_{\alpha\bar{\beta}} = \partial_{\alpha} \partial_{\bar{\beta}} K(z, \bar{z}) \quad K: \text{Kähler potential}$$

$$\frac{\partial}{\partial \bar{z}^{\alpha}} g_{\alpha\bar{\beta}} = \frac{\partial}{\partial \bar{z}^{\alpha}} g_{\bar{\beta}\alpha}(z, \bar{z})$$

$$\Rightarrow g_{\alpha\bar{\beta}}(z, \bar{z}) = \partial_{\alpha} K_{\bar{\beta}}(z, \bar{z})$$

$$\partial_{\bar{x}} (\partial_{\alpha} K_{\bar{\beta}}) = \partial_{\alpha} \partial_{\bar{x}} K_{\bar{\beta}}(z, \bar{z})$$

$$\text{Note that } \bar{g}_{\alpha\bar{\beta}} = g_{\bar{\beta}\alpha} = \partial_{\bar{\beta}} K_{\alpha} = \partial_{\alpha} K_{\bar{\beta}}$$

$$\Rightarrow K_{\alpha} \sim \partial_{\alpha} K$$

$$\text{and so } g_{\alpha\bar{\beta}} = \partial_{\alpha} \partial_{\bar{\beta}} K$$

$$* \quad g_{\alpha\bar{\beta}} = \left( \frac{1}{1+\bar{z}z} \right) \left( \delta_{\alpha\bar{\beta}} - \frac{\bar{z}_{\alpha} z_{\bar{\beta}}}{(1+\bar{z}z)} \right)$$

$$\text{Has Kähler Potential } K = \ln(1+\bar{z}z)$$

and Curvature

$$R_{\alpha\bar{\beta}} = (n+1) g_{\alpha\bar{\beta}}$$

(Kähler-Einstein)

$$\mathbb{CP}^1 \cong S^2: \quad n=1:$$

$$\delta_{\bar{z}z} = 1$$

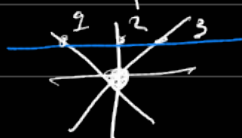
$$g_{\bar{z}z} = \left( \frac{1}{1+\bar{z}z} \right) \left( \delta_{\bar{z}z} - \frac{\bar{z}z}{(1+\bar{z}z)} \right)$$

$$= \frac{1}{(1+\bar{z}z)^2}$$

$$ds^2 = \frac{2}{(1+\bar{z}z)^2} dz d\bar{z}$$

\* Look at  $n$  pts on a line (projective line)

$\mathbb{RP}^1$



$$\bar{x} \sim t \bar{x}$$

$$t \in \mathbb{R}$$

• Parallel line  $\leadsto$  "∞" : 3 pts

$$\bar{x} \sim L \bar{x}$$

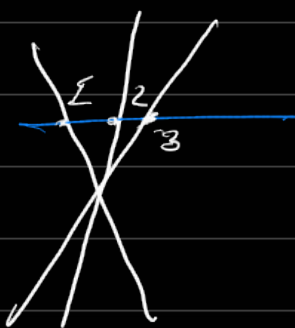
$$\hookrightarrow SL(2)$$

$$\{0, 1, \infty\}$$

$$0: 1$$

$$\infty: \infty$$

$$1: 45^\circ$$



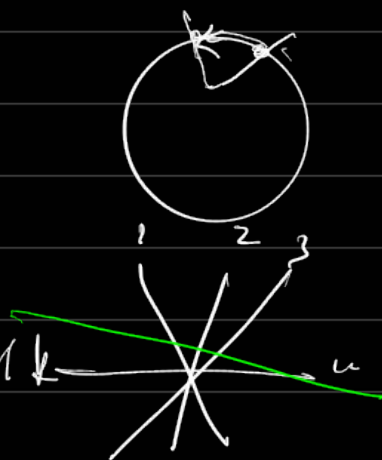
•  $RP^2$  has a natural ordering: 1, 2, 3, ... and each line does not cross.

• Plane  $\cong S^2$ ,

• Concident lines near going to boundary

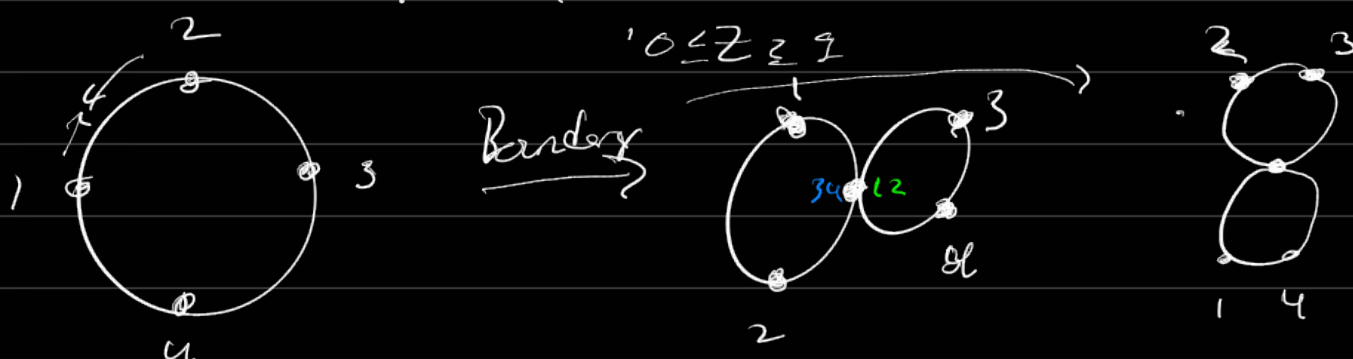
4 points: some D.O.F.

• From  $SL(2)$  we can only get increments from det.



$$\langle ij \rangle := \det(\vec{x}_i, \vec{x}_j)$$

•  $Z = \frac{\langle 12 \rangle \langle 34 \rangle}{\langle 13 \rangle \langle 24 \rangle}$  | Space of 4 pts where 1, ... 4 are ordered is where



Idea: from the pov of  $\langle 12 \rangle$  it looks like  $\langle 34 \rangle$  are squashed together and from the pov of  $\langle 34 \rangle$  it looks

$$Z \rightarrow 0 \Rightarrow \langle 12 \rangle = 0 \quad \text{or} \quad \langle 34 \rangle = 0$$

are when the lines come together  
 $\Rightarrow$  String world sheet.

Schwarzschild tensor:

$$P_{mn} := \frac{1}{2} (R_{mn} - \frac{1}{6} R g_{mn}) \quad \Bigg| \quad \tilde{P} = \tilde{P}(\tilde{g})$$

• Here  $\tilde{P}$ : physical

$$\begin{aligned} \tilde{P}_{mn} &= P_{mn} - \nabla_m \gamma_n + \gamma_m \gamma_n - \frac{1}{2} (\gamma_\rho \gamma^\rho) g_{mn} \\ \gamma_m &= -\Omega \nabla_m \Omega \\ &\Rightarrow \tilde{P}_{mn} = P_{mn} + \Omega^{-1} \nabla_m \nabla_n \Omega - \frac{1}{2} \Omega^{-2} (\nabla_\rho \Omega \nabla^\rho \Omega) g_{mn} \end{aligned}$$

• Einstein eq. on  $\tilde{g}$  means  $\tilde{P}$  is finite as  $\Omega \rightarrow 0$

$$\nabla_\rho \Omega \nabla^\rho \Omega |_x = \eta^\nu \eta^\mu g_{\mu\nu} |_x = 0$$

$$\Rightarrow \nabla_\mu \nabla_\nu \Omega \propto g_{\mu\nu} |_x$$

\* Prop: One can always choose  $(U, z, \bar{z})$  on  $\mathcal{I}^+$   
 weyl freedom  $(n, h) \sim (\lambda^{-1} n, \lambda^2 h)$   
 $n = 2dU, \quad h = 0 \cdot dU^2 + 4dzd\bar{z} \over (1 + |z|^2)^2$

BMS:

Prop (Geroch 77')

The Lie Alg. of int. symm of the universal structure

$$(\mathcal{L}_X n)^a = -\alpha n^a$$

$$(\mathcal{L}_X h)_{ab} = 2\alpha h_{ab} \quad \text{for some } \alpha$$

is generated by

$$X = \left[ \xi(z, \bar{z}) + \frac{U}{z} (\nabla_z X^z + \nabla_{\bar{z}} X^{\bar{z}}) \right] \frac{\partial}{\partial U} + X(z) \partial_z + \bar{X}(\bar{z}) \partial_{\bar{z}}$$

\* This is (11.1.25) with  $(f, \theta) \rightarrow (z, \bar{z})$   
 and  $f \rightarrow \xi, \quad \xi \rightarrow X$

$\Rightarrow \xi(z, \bar{z})$  projective  $S^2$  function

$X$ : conformal vector field on  $S^2$

$$X = a + bz + c\bar{z}^2$$

The Lie Alg. is isomorphic to

$$\text{Lie}(SO(3,1)) \ltimes C^\infty(S^2)$$

(BMS Lie Algebra)

$\Delta$   $SO(3,1)$  acts on a conformal primary of weight of -1

Proof: the expression for  $X$  is given by direct computation in  $(U, z, \bar{z})$

\* If we note  $X_\xi = \xi(z, \bar{z}) \partial_U$  and  $\rho_X = \frac{U}{z} \partial_z + \frac{U}{\bar{z}} \partial_{\bar{z}}$   
 then

$$[X_{\xi_1}, X_{\xi_2}] = 0 \Rightarrow \text{Abelian sub alg} \cong C^\infty(S^2)$$

$\Gamma_U \propto \frac{1}{U}$

$$[X_1, X_2] = X_3, \partial_2 X_2 - X_2 \partial_1 X_1 \quad \text{Lie}(SO(3,1))$$

$$[X_1, X_3] = X_2, \quad \hat{X}_3 = (\chi \partial_2 + \bar{\chi} \partial_{\bar{2}}) - \frac{1}{2} (\partial_2 \chi + \partial_{\bar{2}} \bar{\chi}) \} \quad \text{Cartan primary of C-17}$$

→ If. dim extension of Poincare group

$$\text{Poincare: } \leq SO(3,1) \times \mathbb{R}^4$$

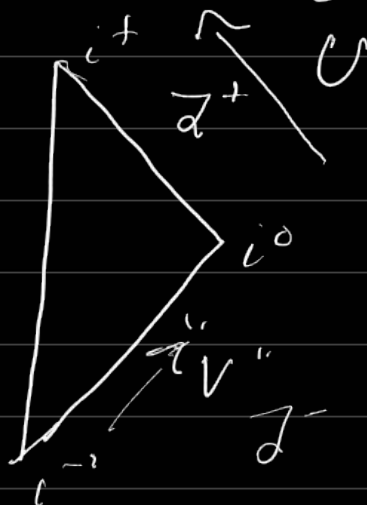
$$\text{BMS: } \text{Lie}(SO(3,1)) \times C^\infty(S^2)$$

•  $X_3$ : translations along  $U$  (along  $\partial/\partial u$ )

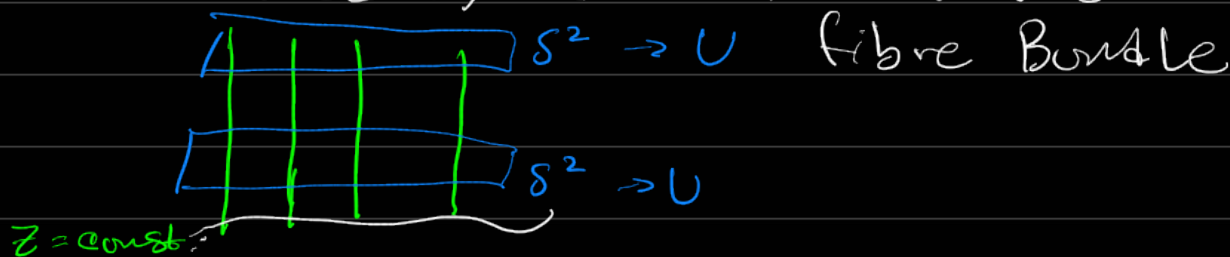
$Y$ : Rotations

↳ look at spherical  
harmonics on  $C^\infty(S^2)$

$$\mathcal{I}^+ \cong S^2 \times \mathbb{R}$$



\*  $\mathcal{I}^+$  is fibred by constant  $Z$  values



• We can Quotient along each fibre to get  $S^2$  with metric

• Each fibre is distinct

Penrose Peeling: Weyl tensor is not invariant.

